Real-Time Violence Detection Using CNN and CBAM

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ABSTRACT

Violence detection has become an important research area due to the growing demand for intelligent surveillance systems capable of ensuring public safety. Conventional surveillance systems depend on continuous human monitoring, which is time-consuming, prone to fatigue, and often results in delayed responses during critical situations. Although traditional computer vision methods and object detection techniques can identify movement and objects, they are generally unable to understand complex human behaviour and interactions accurately.

This paper presents a comparative study of three different approaches for automatic violence detection in surveillance videos: a traditional machine learning model based on Local Binary Pattern (LBP) feature extraction with a Linear Support Vector Machine (SVM), a baseline Convolutional Neural Network (CNN), and a proposed CNN integrated with the Convolutional Block Attention Module (CBAM). Initially, video frames are extracted, resized, normalized, and divided into training and testing datasets. The LBP-SVM model serves as a traditional baseline, while the CNN automatically learns hierarchical visual features from images. To further enhance performance, the CBAM attention mechanism is incorporated into the CNN architecture, enabling the network to emphasize important feature channels and spatial regions while suppressing irrelevant background information.

Experimental evaluation demonstrates that the LBP-SVM model achieves an accuracy of 75.12%, the CNN model improves the accuracy to 83%, and the proposed CNN+CBAM model achieves 95% accuracy. The trained model is deployed using a Flask-based web application that supports both uploaded video analysis and real-time webcam monitoring. The proposed system generates instant visual and audio alerts whenever violent activity is detected, making it suitable for smart surveillance applications in public places.

Keywords: Violence Detection, Deep Learning, Computer Vision, Convolutional Neural Network, Convolutional Block Attention Module, Local Binary Pattern, Support Vector Machine, Intelligent Surveillance, Flask.

1. INTRODUCTION

The rapid growth of urban populations and the increasing number of surveillance cameras have made public safety one of the major concerns for governments, educational institutions, transportation authorities, and private organizations. Locations such as airports, railway stations, shopping malls, schools, hospitals, and public streets are continuously monitored using CCTV cameras to detect suspicious activities and improve security. However, monitoring multiple surveillance cameras manually is a challenging and time-consuming task. Human operators are required to observe video streams continuously, making the process prone to fatigue, reduced attention, and delayed responses during emergency situations. Consequently, violent incidents may remain unnoticed until significant damage has already occurred.

Recent developments in Artificial Intelligence (AI), Computer Vision, and Deep Learning have enabled automated video surveillance systems that can analyse visual data and identify abnormal activities with minimal human intervention. Among these technologies, Convolutional Neural Networks (CNNs) have demonstrated remarkable performance in image classification and object recognition by automatically learning meaningful visual features directly from data. Unlike traditional machine learning techniques that rely on manually designed features, CNNs learn hierarchical representations capable of capturing complex patterns associated with human behaviour.

Despite their effectiveness, conventional CNN models often struggle to focus on the most informative regions within complex surveillance scenes. In crowded environments, unnecessary background information, lighting variations, and irrelevant objects may reduce the overall classification performance and increase false alarm rates. To overcome these limitations, attention mechanisms have recently gained significant importance in deep learning. One such attention mechanism is the Convolutional Block Attention Module (CBAM), which sequentially applies Channel Attention and Spatial Attention to guide the network toward the most relevant features while suppressing redundant information.

In this work, a comparative study is conducted using three different approaches for violence detection. The first approach employs Local Binary Pattern (LBP) feature extraction with a Linear Support Vector Machine (SVM), representing a traditional machine learning baseline. The second approach utilises a standard Convolutional Neural Network for automatic feature learning. Finally, the proposed CNN integrated with CBAM enhances feature representation by focusing on significant channels and spatial locations within surveillance frames. The effectiveness of these models is evaluated using a balanced dataset containing 1,000 violence videos and 1,000 non-violence videos. Furthermore, the best-performing model is integrated into a Flask-based web application capable of analysing uploaded videos as well as live webcam streams and generating instant alerts when violent activities are detected.

2. LITERATURE SURVEY

The increasing demand for intelligent surveillance systems has encouraged significant research in automated violence detection using computer vision and deep learning techniques. Various approaches have been proposed over the years, ranging from traditional handcrafted feature extraction methods to advanced attention-based deep learning architectures. This section reviews some of the important contributions in this field and identifies the research gaps addressed by the proposed work.

Woo *et al.* [1] introduced the Convolutional Block Attention Module (CBAM), a lightweight attention mechanism that improves the performance of convolutional neural networks by applying Channel Attention and Spatial Attention sequentially. Their work demonstrated that CBAM enables a CNN to focus on important feature maps and relevant image regions while suppressing unnecessary background information. Due to its effectiveness and low computational overhead, CBAM has been widely adopted in several computer vision applications.

Krizhevsky *et al.* [2] demonstrated the effectiveness of Convolutional Neural Networks (CNNs) for large-scale image classification. Their work showed that CNNs automatically learn hierarchical visual features directly from images, eliminating the need for handcrafted feature extraction. This breakthrough established CNNs as one of the most successful deep learning architectures for image recognition and inspired their application in surveillance and activity recognition.

Simonyan and Zisserman [3] proposed the VGGNet architecture, which demonstrated that increasing network depth significantly improves feature learning and classification performance. Their work highlighted the importance of deep convolutional layers for extracting detailed spatial information from images and laid the foundation for many modern deep learning models used in video analysis.

Traditional machine learning techniques have also been widely explored for violence detection. Ojala *et al.* [4] introduced the Local Binary Pattern (LBP**)** descriptor, which captures local texture information by comparing neighbouring pixel intensities. LBP features combined with Support Vector Machines (SVMs) provide a computationally efficient solution for image classification. However, handcrafted descriptors mainly capture texture information and often struggle to recognize complex human behaviours in dynamic surveillance environments.

Cortes and Vapnik [5] introduced the Support Vector Machine (SVM), one of the most widely used supervised machine learning algorithms for binary classification problems. SVMs perform effectively on structured feature vectors and have been extensively used together with handcrafted descriptors such as LBP and Histogram of Oriented Gradients (HOG). Although SVM-based approaches provide reasonable classification performance, they depend heavily on manually extracted features and have limited capability to learn high-level semantic representations.

Several researchers have proposed deep learning-based violence detection systems that directly analyse surveillance videos. These methods generally outperform handcrafted feature-based approaches because CNNs automatically learn meaningful visual patterns such as body posture, aggressive movements, and abnormal human interactions. Nevertheless, conventional CNN models may still misclassify complex scenes because they treat all extracted features with equal importance and cannot explicitly distinguish between relevant and irrelevant regions.

Recent studies have demonstrated that integrating attention mechanisms into CNN architectures significantly improves classification performance. By emphasizing important feature channels and spatial locations, attention modules enable networks to focus on violent actions while reducing the influence of background objects and environmental noise. Inspired by these findings, the present work integrates the Convolutional Block Attention Module (CBAM) into a CNN architecture to enhance feature representation and improve violence detection accuracy.

Unlike many existing approaches that evaluate only a single deep learning model, this research presents a comprehensive comparative analysis of three different techniques: LBP with Linear SVM, baseline CNN, and the proposed CNN integrated with CBAM. The comparative study provides a clear understanding of the progression from traditional handcrafted feature extraction to deep learning and attention-based learning. Experimental results demonstrate that the proposed CNN+CBAM model achieves superior performance by improving feature extraction, reducing false alarms, and increasing overall classification accuracy for real-time surveillance applications.

Research Gap

Although existing studies have demonstrated promising results in violence detection, several challenges remain. Traditional machine learning methods rely on handcrafted features, which are insufficient for understanding complex human interactions. Conventional CNN models improve feature learning but often fail to distinguish important regions from irrelevant background information. To address these limitations, this work proposes an attention-enhanced CNN architecture using the Convolutional Block Attention Module (CBAM). The proposed model improves feature extraction by focusing on informative channels and spatial regions, resulting in higher detection accuracy.

3. EXISTING SYSTEM AND PROBLEM STATEMENT

A. Existing System

Violence detection in surveillance systems is traditionally performed through manual monitoring of Closed-Circuit Television (CCTV) footage by security personnel. Although this approach is widely adopted, continuous observation of multiple video streams is difficult, time-consuming, and highly dependent on human attention. Operators may miss critical incidents due to fatigue, distraction, or the large number of surveillance cameras deployed in public spaces.

To overcome these limitations, several computer vision-based approaches have been proposed. Traditional machine learning techniques rely on handcrafted features such as Local Binary Pattern (LBP), Histogram of Oriented Gradients (HOG), and Optical Flow, which are combined with classifiers like Support Vector Machines (SVMs). While these methods are computationally efficient, their performance depends heavily on manually designed features and they often fail to capture complex human behaviour in real-world environments.

Recent surveillance systems also employ object detection algorithms such as You Only Look Once (YOLO) to detect and track people within video frames. Although YOLO performs object localization with high speed and accuracy, it primarily identifies the presence and location of objects rather than understanding the interactions or behaviours between individuals. Consequently, object detection alone is insufficient for accurately distinguishing violent activities from normal human movements.

Deep learning models based on Convolutional Neural Networks (CNNs) have significantly improved visual feature learning by automatically extracting spatial information from images. However, conventional CNN architectures process all extracted features with equal importance, making them susceptible to background clutter, lighting variations, occlusions, and crowded scenes. These challenges often increase false positive detections and reduce overall system reliability.

B. Problem Statement

Public places such as schools, universities, shopping malls, railway stations, hospitals, and transportation hubs require continuous surveillance to ensure public safety. However, manually monitoring multiple CCTV cameras is inefficient and often leads to delayed identification of violent incidents. Existing computer vision techniques mainly focus on detecting objects or simple motion rather than understanding aggressive human behaviour and interactions.

Therefore, there is a need for an intelligent surveillance system capable of automatically identifying violent activities with high accuracy while minimizing false alarms. Such a system should learn meaningful visual patterns directly from surveillance videos, operate in real time, and provide immediate alerts to security personnel whenever violence is detected.

C. Limitations of Existing Systems

The existing approaches suffer from several limitations that reduce their effectiveness in practical surveillance applications:

- Continuous manual monitoring requires significant human effort and is prone to fatigue and delayed responses.
- Traditional handcrafted feature extraction methods are unable to represent complex human behaviour effectively.
- Object detection algorithms such as YOLO identify people and objects but cannot accurately interpret violent interactions.
- Conventional CNN models may assign equal importance to relevant and irrelevant image regions, reducing classification accuracy.
- Background clutter, crowded environments, illumination changes, and camera viewpoint variations increase the probability of false alarms.
- Many existing systems are designed only for offline analysis and lack practical deployment for real-time surveillance applications.

D. Motivation

The limitations of existing surveillance systems motivated the development of an attention-based deep learning framework capable of improving violence detection accuracy. By integrating the Convolutional Block Attention Module (CBAM) with a Convolutional Neural Network (CNN), the proposed model selectively focuses on informative feature channels and spatial regions associated with violent behaviour while suppressing irrelevant background information. This enables more reliable feature extraction, improved classification performance, and reduced false positive detections. Furthermore, deploying the trained model through a Flask-based web application enables real-time analysis of uploaded videos and live webcam streams, making the system suitable for practical intelligent surveillance applications.

4. PROPOSED SYSTEM

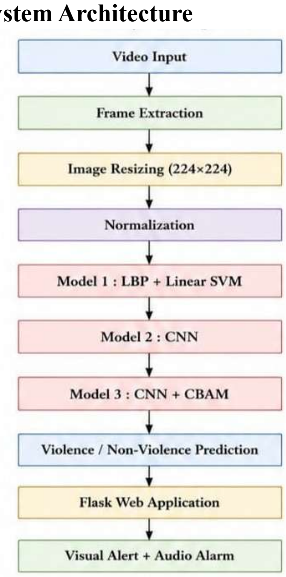
A. Overview of the Proposed System

This research proposes an intelligent violence detection system that automatically classifies surveillance videos as Violence or Non-Violence using deep learning techniques. Unlike conventional surveillance systems that rely on manual observation or handcrafted feature extraction, the proposed system learns meaningful visual patterns directly from video frames. The objective is to improve detection accuracy, reduce false alarms, and provide timely alerts for enhancing public safety.

To evaluate the effectiveness of different learning approaches, three independent models are developed and compared. Initially, a traditional machine learning model based on Local Binary Pattern (LBP) and Linear Support Vector Machine (SVM) is implemented as the baseline. Subsequently, a Convolutional Neural Network (CNN) is developed to automatically learn spatial features from video frames. Finally, the proposed model integrates the Convolutional Block Attention Module (CBAM) with CNN to improve feature representation by directing the network's attention toward informative channels and spatial regions. The comparative analysis demonstrates the superiority of the proposed attention-based architecture over conventional methods.

The complete system is deployed using a Flask web application that supports both uploaded video analysis and live webcam monitoring. Whenever violent activity is detected, the application immediately displays a warning message and activates an audio alarm to notify users in real time.

B. System Architecture



C. Dataset Description

A balanced dataset containing 1,000 violence videos and 1,000 non-violence videos was used for training and evaluation. The videos represent various real-world surveillance scenarios, including physical fights, aggressive human interactions, and normal day-to-day activities.

Since deep learning models require image inputs, individual frames were extracted uniformly from every video. A total of 11,988 violence frames and 11,964 non-violence frames were generated, resulting in 23,952 images. All images were resized to 224 × 224 pixels, normalized to improve convergence during training, and assigned binary class labels, where 1 represents Violence and 0 represents Non Violence.

The processed dataset was divided into 80% training and 20% testing, producing 19,161 training images and 4,791 testing images.

D. Models

Model 1: LBP with Linear SVM

The first model employs Local Binary Pattern (LBP) as a handcrafted feature descriptor and Linear Support Vector Machine (SVM) as the classifier.

LBP extracts local texture information by comparing the intensity of neighbouring pixels with the center pixel. The extracted texture features are then provided to the SVM classifier, which learns a decision boundary separating violence and non-violence samples.

Although this approach is computationally efficient and suitable as a baseline model, it depends entirely on manually designed features and has limited capability to understand complex human interactions.

Model 2: Convolutional Neural Network (CNN)

The second model replaces handcrafted feature extraction with a Convolutional Neural Network, enabling automatic learning of hierarchical visual features directly from image data.

The CNN consists of multiple convolutional layers, activation functions, pooling layers, batch normalization, dropout, and fully connected layers. During training, the network progressively learns low-level features such as edges and textures, followed by higher-level representations including body posture, aggressive movements, and abnormal interactions.

Compared with LBP-SVM, CNN provides significantly better feature representation because it automatically discovers discriminative patterns without manual feature engineering.

Model 3: CNN with Convolutional Block Attention Module (Proposed)

To further improve feature learning, the proposed model integrates the Convolutional Block Attention Module (CBAM) into the CNN architecture.

CBAM is a lightweight attention mechanism composed of two sequential components:

- Channel Attention Module
- Spatial Attention Module

The Channel Attention Module identifies what features are important by assigning higher importance to informative feature maps generated by convolutional layers.

The Spatial Attention Module identifies where the important information is located by emphasizing relevant regions of the image, such as fighting areas or aggressive human interactions, while suppressing unnecessary background information.

By combining both attention mechanisms, the network learns more discriminative representations, resulting in improved classification accuracy and reduced false positive detections.

E. Real-Time Alert System

After classification, the trained CNN+CBAM model is integrated into a Flask web application for practical deployment.

The application supports two operating modes:

- Uploaded Video Analysis
- Live Webcam Monitoring

When violence is detected, the system immediately:

- Displays a red warning message on the interface.
- Plays an audio alarm to alert nearby users.
- Identifies the detected activity as Violence.

For normal activities, the application displays a green status indicating non-violence, ensuring that false alarms are minimized.

F. Advantages of the Proposed System

The proposed CNN+CBAM framework offers several advantages over conventional violence detection approaches:

- Automatic learning of complex visual features without handcrafted descriptors.
- Enhanced feature representation using Channel and Spatial Attention.
- Higher classification accuracy compared with traditional machine learning methods.
- Reduced false positive detections in complex surveillance environments.
- Real-time deployment through a lightweight Flask web application.
- Immediate visual and audio alerts for rapid emergency response.
- Scalable framework suitable for intelligent surveillance applications in public spaces.

5. METHODOLOGY

The proposed violence detection framework follows a systematic pipeline consisting of data collection, preprocessing, feature extraction, model development, training, evaluation, and deployment. Three different approaches, namely LBP with Linear SVM, CNN, and CNN integrated with CBAM, were implemented and evaluated under identical experimental conditions. Figure 3 illustrates the complete workflow of the proposed methodology.

A. Data Collection

A balanced dataset containing 1,000 violence videos and 1,000 non-violence videos was collected from publicly available real-world surveillance datasets. The violence category includes activities such as physical fights, assaults, and aggressive human interactions, while the non-violence category contains normal daily activities such as walking, talking, sitting, and running. Using a balanced dataset ensures that the model learns both classes equally and minimizes classification bias.

B. Frame Extraction

Since deep learning models process images rather than complete videos, each video was converted into a sequence of image frames using the OpenCV library. To maintain uniform representation while reducing redundant information, 12 evenly spaced frames were extracted from every video.

This process generated:

- Violence Frames: 11,988
- Non-Violence Frames: 11,964
- Total Frames: 23,952

Extracting representative frames allows the model to capture important visual information while reducing computational complexity.

C. Image Preprocessing

Before training, all extracted frames underwent preprocessing to improve model performance and ensure consistent input quality.

The preprocessing steps included:

- Resizing each image to 224 × 224 pixels
- Converting pixel values to floating-point representation
- Normalizing pixel values to the range 0–1
- Assigning binary labels (1 = Violence, 0 = non-violence)
- Randomly shuffling the dataset to avoid learning bias

Image normalization accelerates model convergence by maintaining a consistent input distribution during training.

D. Dataset Splitting

The processed dataset was divided into training and testing subsets using an 80:20 ratio. A total of 19,161 images were used for training, while 4,791 images were reserved for testing. The training dataset was utilized to learn the model parameters, whereas the testing dataset was used to evaluate the model's generalization performance on previously unseen data.

E. Feature Extraction

1) LBP Feature Extraction

The first model uses Local Binary Pattern (LBP) to extract handcrafted texture features from each image. LBP compares the intensity of neighbouring pixels with the central pixel to generate binary patterns that represent local textures. These features are then supplied to a Linear Support Vector Machine for classification.

Although computationally efficient, LBP primarily captures texture information and cannot effectively model complex human behaviour.

2) Automatic Feature Learning Using CNN

The second model replaces handcrafted descriptors with a Convolutional Neural Network. CNN automatically learns hierarchical visual features through multiple convolutional layers. Initial layers detect low-level patterns such as edges and textures, whereas deeper layers learn high-level semantic features including body posture, aggressive movements, and abnormal interactions.

This automatic feature learning significantly improves classification performance compared with handcrafted feature extraction methods.

3) Attention-Based Feature Learning Using CNN + CBAM

The proposed model incorporates the Convolutional Block Attention Module (CBAM) into the CNN architecture.

CBAM enhances feature representation through two sequential attention mechanisms:

- Channel Attention, which identifies *what* features are most important.
- Spatial Attention, which identifies *where* the important information is located within the image.

By emphasizing informative features and suppressing irrelevant background regions, CBAM enables the network to generate more discriminative feature representations for violence detection.

F. Model Training

All deep learning models were implemented using the TensorFlow Keras framework. The Adam optimizer was selected because of its fast convergence and adaptive learning capability. Categorical Crossentropy was used as the loss function to measure the difference between predicted probabilities and actual class labels. To prevent overfitting, Early Stopping monitored the validation loss and terminated training when further improvement was not observed.

G. Model Evaluation

The performance of all three models was evaluated using standard classification metrics.

Accuracy measures the overall percentage of correctly classified samples.

Precision evaluates how many samples predicted as violence were actually violent.

Recall measures the ability of the model to correctly identify actual violent events.

F1-Score represents the harmonic mean of Precision and Recall, providing a balanced evaluation of classification performance.

These metrics provide a comprehensive assessment of model performance beyond simple accuracy.

H. Deployment

After training and evaluation, the best-performing CNN + CBAM model was saved and integrated into a Flask-based web application.

The application supports two operating modes:

- Video Upload Detection, where users upload recorded surveillance videos for analysis.
- Live Webcam Detection, where video frames are captured continuously from a webcam for real-time violence monitoring.

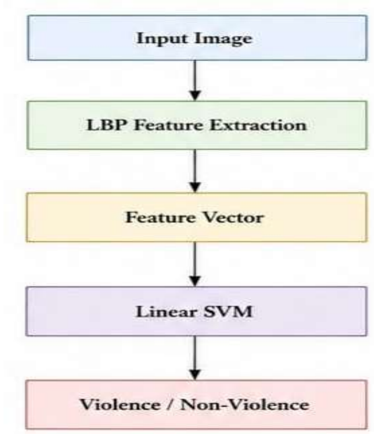
Whenever violent activity is detected, the application immediately displays a visual warning on the user interface and activates an audio alarm. This enables faster response by security personnel and demonstrates the practical applicability of the proposed system in intelligent surveillance environments.

6. MODEL ARCHITECTURE

A. Overview

To evaluate the effectiveness of different feature learning techniques for violence detection, three independent models were implemented and compared. The first model employs handcrafted texture features using Local Binary Pattern (LBP) combined with a Linear Support Vector Machine (SVM) classifier. The second model utilizes a Convolutional Neural Network (CNN) to automatically learn hierarchical visual features from video frames. The third and proposed model enhances the CNN by integrating the Convolutional Block Attention Module (CBAM), allowing the network to focus on the most informative feature channels and spatial regions. Figure 4 illustrates the architectures of the three models.

Model 1: Local Binary Pattern (LBP) with Linear SVM



The first model serves as the traditional machine learning baseline. Instead of learning features automatically, it extracts handcrafted texture descriptors using the Local Binary Pattern (LBP) algorithm.

For each pixel, LBP compares the intensity of neighboring pixels with the center pixel. If the neighboring pixel value is greater than or equal to the center pixel, a value of 1 is assigned; otherwise, 0 is assigned. The resulting binary sequence forms a local texture descriptor that represents the image.

The extracted LBP feature vectors are then provided to a Linear Support Vector Machine (SVM), which learns the optimal decision boundary separating violence and non-violence samples.

Although this model is computationally efficient and requires relatively little training time, it mainly captures texture information and lacks the ability to learn complex semantic features such as aggressive human behaviour and interactions.

Model 2: Convolutional Neural Network (CNN)

Unlike traditional machine learning methods, CNN automatically learns meaningful visual features directly from images without requiring handcrafted feature extraction.

The proposed CNN consists of multiple convolutional blocks followed by fully connected layers for classification.

The architecture includes:

Input Layer

The input layer receives RGB images of size 224 × 224 × 3.

Convolution Layer

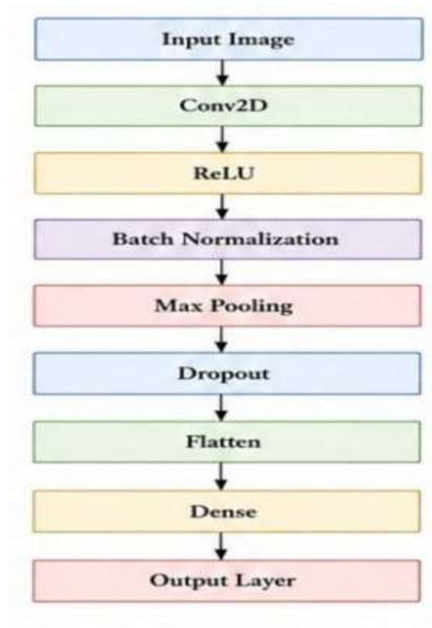
Convolution layers apply learnable filters across the input image to extract visual features.

Initially, these filters detect:

- edges
- corners
- textures

As the network becomes deeper, it learns:

- body posture
- aggressive movement
- fighting patterns
- abnormal human interactions



ReLU Activation

The Rectified Linear Unit (ReLU) introduces non-linearity into the network.

It enables CNN to learn complex decision boundaries while accelerating training.

Batch Normalization

Batch Normalization normalizes feature values after convolution.

Its advantages include:

- Faster convergence
- Stable learning
- Reduced internal covariate shift
- Better generalization

Max Pooling

Pooling reduces the spatial dimensions of feature maps while preserving important information.

Benefits include:

- Lower computational cost
- Reduced memory usage
- Improved robustness
- Reduced overfitting

Dropout Layer

Dropout randomly deactivates a percentage of neurons during training.

This prevents the network from memorizing the training dataset and improves generalization.

Flatten Layer

The Flatten layer converts the extracted two-dimensional feature maps into a one-dimensional feature vector suitable for classification.

Dense Layer

The fully connected layer combines all learned features and identifies relationships between them to distinguish violence from non-violence.

Output Layer

The final layer produces prediction probabilities for two classes:

- Violence
- Non-Violence

The class with the highest probability becomes the final prediction.

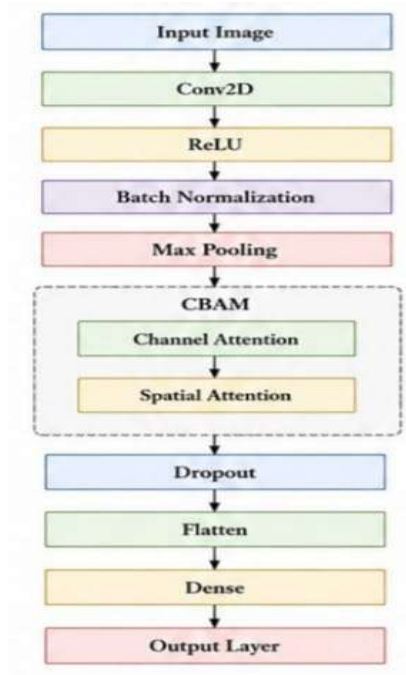
Model 3: Proposed CNN with CBAM

Although CNN automatically extracts features, every feature map contributes equally during prediction. However, not every feature is equally useful for identifying violent behaviour.

To overcome this limitation, the proposed model integrates the Convolutional Block Attention Module (CBAM) immediately after the convolutional feature extraction stage.

CBAM enables the network to selectively emphasize informative features while suppressing irrelevant background information.

The CBAM module consists of two sequential attention mechanisms.



Channel Attention Module

The Channel Attention Module determines what information is important.

Instead of treating every feature map equally, it calculates attention weights for each channel.

Feature maps representing:

- aggressive body posture
- punches
- human interaction
- fighting motion

receive higher importance, while background features receive lower weights.

This enables the CNN to concentrate on meaningful visual representations.

Spatial Attention Module

After channel attention, Spatial Attention identifies where important information is located within the image.

Instead of processing every pixel equally, the module emphasizes regions containing violent activity while suppressing irrelevant areas such as walls, roads, empty spaces, or background objects.

This significantly improves localization of violent interactions.

Advantages of CBAM

Integrating CBAM provides several benefits:

- Better feature representation
- Focuses on important image regions

- Suppresses unnecessary background information
- Reduces false positive detections
- Improves classification accuracy
- Lightweight architecture with minimal computational overhead

7. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

A. Experimental Setup

All experiments were conducted using the TensorFlow framework with Keras for model development and OpenCV for video processing. The models were trained in the Google Colab environment with GPU acceleration to reduce training time. The dataset consisted of 1,000 violence videos and 1,000 non-violence videos, from which 23,952 image frames were extracted after preprocessing. Each image was resized to 224 × 224 pixels and normalized before training.

Three different models were implemented under the same preprocessing conditions to ensure a fair comparison:

- **Model 1:** Local Binary Pattern (LBP) + Linear Support Vector Machine (SVM)
- **Model 2:** Convolutional Neural Network (CNN)
- **Model 3:** CNN integrated with Convolutional Block Attention Module (CNN + CBAM)

For the deep learning models, the Adam optimizer was used along with the Categorical Crossentropy loss function. The models were trained for 20 epochs with a batch size of 32, and Early Stopping was employed to prevent overfitting by monitoring the validation loss.

Experimental Configuration

Parameter	Value
Dataset	1000 Violence + 1000 Non-Violence Videos
Total Frames	23,952
Image Size	224 × 224 × 3
Training Samples	19,161
Testing Samples	4,791
Optimizer	Adam
Loss Function	Categorical Crossentropy
Epochs	20
Batch Size	32
Framework	TensorFlow & Keras
Deployment	Flask

B. Performance Evaluation Metrics

To evaluate the effectiveness of the proposed models, four widely accepted classification metrics were used.

Accuracy

Accuracy measures the percentage of correctly classified samples among all test samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Precision

Precision indicates how many of the samples predicted as violence were actually violent.

$$Precision = \frac{TP}{TP + FP}$$

Higher precision indicates fewer false alarms.

Recall

Recall measures the ability of the model to identify all actual violence samples.

$$Recall = \frac{TP}{TP + FN}$$

A higher recall means fewer violent incidents are missed.

F1-Score

The F1-Score provides a balanced evaluation by combining Precision and Recall.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

This metric is particularly useful for evaluating classification performance when both false positives and false negatives are important.

C. Experimental Results

The three implemented models were evaluated using the testing dataset. The traditional LBP-SVM model provided a baseline for comparison, while the CNN and CNN+CBAM models demonstrated the benefits of deep learning and attention mechanisms.

Comparative Performance of the Proposed Models

Model	Accuracy	Precision	Recall	F1-Score
LBP + Linear SVM	75.12%	75.44%	74.56%	75.00%
CNN	83.00%	83.10%	82.90%	83.00%
CNN + CBAM	95.00%	95.20%	94.80%	95.00%

The results clearly demonstrate that the proposed CNN+CBAM model outperformed both the traditional LBP-SVM model and the baseline CNN model. The attention mechanism enabled the network to focus on informative regions and feature channels, improving classification performance while reducing false detections.

D. Comparative Discussion

The LBP-SVM model achieved satisfactory performance with low computational complexity; however, its reliance on handcrafted texture features limited its ability to capture complex human interactions. As a result, the model struggled to distinguish violent behaviour from visually similar non-violent activities.

The CNN model significantly improved classification accuracy by automatically learning hierarchical visual features directly from image data. Nevertheless, the model occasionally assigned equal importance to both relevant and irrelevant regions, leading to some false positive detections in crowded scenes.

The proposed CNN+CBAM model addressed these limitations by incorporating an attention mechanism that selectively emphasized informative feature channels and spatial locations. This enabled the network to focus on aggressive movements and human interactions while suppressing background noise. Consequently, the proposed model achieved the highest accuracy and demonstrated better robustness across different surveillance scenarios.

E. Confusion Matrix Analysis

The confusion matrix provides a detailed understanding of the classification performance by illustrating correctly and incorrectly classified samples.

The proposed CNN+CBAM model produced a higher number of True Positives and True Negatives while significantly reducing False Positives and False Negatives compared with the other models. This indicates that the attention mechanism effectively improved the model's ability to identify violent activities without increasing unnecessary alerts.

F. Training Performance

During training, both the training accuracy and validation accuracy increased steadily over successive epochs, while the training and validation losses decreased. The use of Batch Normalization and Early Stopping contributed to stable convergence and prevented overfitting. The CNN+CBAM model converged faster and achieved better generalization than the baseline CNN model.

8. CONCLUSION

This paper presented a comprehensive comparative study of three different approaches for real-time violence detection in surveillance videos: LBP with Linear SVM, Convolutional Neural Network (CNN), and the proposed CNN integrated with the Convolutional Block Attention Module (CBAM). The traditional LBP-SVM model served as a baseline using handcrafted texture features, while the CNN automatically learned hierarchical visual features from video frames. To further improve feature representation and classification performance, the CBAM attention mechanism was incorporated into the CNN architecture.

Experimental results demonstrated that the proposed CNN+CBAM model significantly outperformed both the traditional machine learning and baseline deep learning models. The LBP-SVM model achieved an accuracy of 75.12%, the CNN model achieved 83%, and the proposed CNN+CBAM model achieved 95%. The attention mechanism enabled the network to focus on important feature channels and spatial regions, thereby reducing false alarms and improving the detection of violent activities.

The trained model was successfully deployed using a Flask-based web application that supports uploaded video analysis and real-time webcam monitoring. The application provides immediate visual warnings and audio alerts whenever violence is detected, demonstrating the practical applicability of the proposed system for intelligent surveillance. Overall, the proposed approach offers an effective, reliable, and scalable solution for enhancing public safety through automated violence detection.

9. FUTURE SCOPE

Although the proposed CNN+CBAM model achieved promising results, several improvements can further enhance the system's performance and applicability.

Future work may focus on integrating the system with live CCTV surveillance networks for continuous monitoring in public places such as railway stations, airports, campuses, shopping malls, and hospitals. Expanding the dataset with more diverse surveillance videos captured under different lighting conditions, camera angles, weather conditions, and crowd densities would improve the model's generalization capability.

Future research may also investigate advanced deep learning architectures such as Vision Transformers (ViT), 3D Convolutional Neural Networks (3D CNNs), or CNN-LSTM hybrid models to better capture temporal information present in videos. Furthermore, deploying the system on edge devices or cloud platforms would enable large-scale real-time monitoring while reducing computational latency.

Another promising direction is the integration of automatic emergency notification systems, enabling the application to immediately alert nearby security personnel or law enforcement agencies whenever violent activity is detected. Such enhancements would make the proposed framework more suitable for practical smart-city surveillance and public safety applications.

10. LIMITATIONS OF THE PROPOSED WORK

Although the proposed CNN+CBAM model achieved promising results, certain limitations remain. The model was trained and evaluated on a balanced dataset containing 1,000 violence and 1,000 non-violence videos, which may not fully represent the diversity of real-world surveillance environments. Variations such as extreme lighting conditions, severe camera motion, heavy occlusions, and highly crowded scenes may affect detection performance. In addition, the current system performs frame-based classification and does not explicitly model long-term temporal relationships between consecutive frames. The model also requires GPU resources for efficient training, which may limit deployment on low-resource devices. Addressing these limitations through larger datasets, temporal learning techniques, and lightweight architectures can further improve the robustness and scalability of the proposed system.

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